Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following represents a redox reaction?

(A) $NaOH + HCl \longrightarrow NaCl + H_2O$

(B) $BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 + 2HCl$

(C) $CuSO_4 + 2H_2O \longrightarrow Cu(OH)_2 + H_2SO_4$

(D) $Zn + 2HCl \longrightarrow ZnCl_2 + H_2$

Q.2

Zn gives H_2 gas with H_2SO_4 and HCl but not with HNO_3 because

(A) Zn acts as an oxidising agent when it reacts with HNO_3

(B) HNO_3 is weaker acid than H_2SO_4 and HCl

(C) In electrochemical series, Zn is above hydrogen

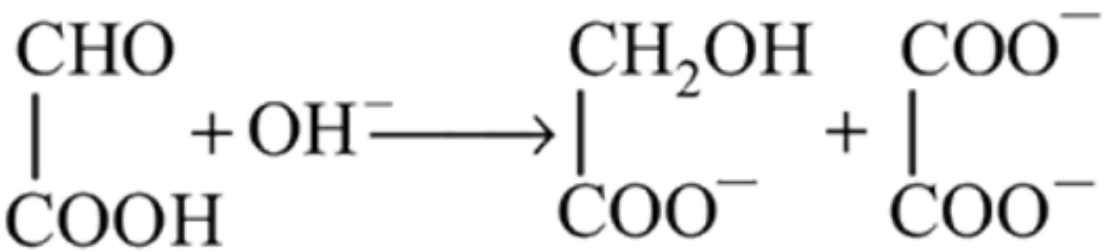
(D) NO_3^- is reduced in preference to hydronium ion

Q.3

Which of the following is not disproportionation reaction?

(A) $2PhCHO \xrightarrow{Al(OEt)_3} PhCOOCH_2Ph$

(B)



(C) $NaH + H_2O \longrightarrow NaOH + H_2$

(D) All of the above.

Q.4

Which of the following represents redox reactions?

I. $Cr_2O_7^{2-} + 2\overline{O}H \longrightarrow 2Cr_4^{2-} + H_2O$

II. $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$

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III. $2MnO_4^- + 3Mn^{2+} + 4\overline{O}H \longrightarrow 5MnO_2 + 2H_2O$

IV. $2Cu^+ \longrightarrow Cu + Cu^{2+}$

(A) I,II

(B) I, III

(C) III, IV

(D) II, III,IV

Q.5

In the disproportionation reaction $3HClO_3 \longrightarrow HClO_4 + Cl_2 + 2O_2 + H_2O$, the equivalent mass of the oxidizing agent is (molar mass of $HClO_3 = 84.45$)

(A) 16.89

(B) 32.22

(C) 84.45

(D) 28.15

Q.6

Which of the following is a redox reaction?

(A) $NaCl + KNO_3 \rightarrow NaNO_3 + KCl$

(B) $CaC_2O_4 + 2HCl \rightarrow CaCl_2 + H_2C_2O_4$

(C) $Mg(OH)_2 + 2NH_4Cl \rightarrow MgCl_2 + 2NH_4OH$

(D) $Zn + 2AgCN \rightarrow 2Ag + Zn(CN)_2$

Q.7

Which of the following do not show disproportionation reaction? $ClO_4^-, F_2, Cl_2, ClO_2^-, ClO_2^-, P_4,\ S_8$, and ClO^-

(A) ClO_2^-, ClO_4^-, and ClO^-

(B) F_2 only

(C) F_2 and ClO_4^-

(D) ClO_4^- only

Q.8

Which of the following is not an intermolecular redox reaction?

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(A) $MgCO_3 \longrightarrow MgO + CO_2$

(B) $O_2 + 2H_2 \longrightarrow 2H_2O$

(C) $K + H_2O \longrightarrow KOH + (1/2)H_2$

(D) $MnBr_3 \longrightarrow MnBr_2 + (1/2)Br_2$

Q.9

The oxidation number of S in $Na_2\ S_4O_6$ is

(A) $+0.5$

(B) 2.5

(C) $+4$

(D) +6

Q.10

The oxidation number of Fe in $Na_2\ [Fe(CN)_5NO]$
is

(A) $+2$

(B) $+1$

(C) $+3$

(D) -2

Q.11

The oxidation state of Fe in Fe_3O_8 is

(A) $3/2$

(B) $4/5$

(C) $5/4$

(D) $16/3$

Q.12

The oxidation number of Pt in $[Pt\ (C_2H_4)\ Cl_3]^-$ is :

(A) $+1$

(B) $+2$

(C) $+3$

(D) +4

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Q.13

Which of the following statements are correct concerning redox properties?
(i) A metal M for which E° for the half life reaction $\mathrm{M}^{n+} + \mathrm{ne}^{-} \rightleftharpoons \mathrm{M}$ is very negative will be a good reducing agent.
(ii) The oxidizing power of the halogens decreases from chlorine to iodine.
(iii) The reducing power of hydrogen halides increases from hydrogen chloride to hydrogen iodide

(A) (i), (ii) and (iii)

(B) (i) and (ii)

(C) (i) only

(D) (ii) and (iii)

Q.14

Given:
$\mathrm{XNa_2HAsO_3 + YNaBrO_3 + ZHCl \rightarrow}$
$\mathrm{NaBr + H_3AsO_4 + NaCl}$
The values of X, Y and Z in the above redox reaction are
respectively:

(A) $2, 1, 2$

(B) $2, 1, 3$

(C) $3, 1, 6$

(D) $3, 1, 4$

Q.15

Oxidation state of sulphur in anions $\mathrm{SO_3^{2-}}, \mathrm{S_2O_4^{2-}}$ and $\mathrm{S_2O_6^{2-}}$ increases in the orders:

(A) $\mathrm{S_2O_6^{2-} < S_2O_4^{2-} < SO_3^{2-}}$

(B) $\mathrm{SO_6^{2-} < S_2O_4^{2-} < S_2O_6^{2-}}$

(C) $\mathrm{S_2O_4^{2-} < SO_3^{2-} < S_2O_6^{2-}}$

(D) $\mathrm{S_2O_4^{2-} < S_2O_6^{2-} < SO_3^{2-}}$

Q.16

Amongst the following, identify the species with an atom in $+6$ oxidation state:

(A) $\mathrm{[MnO_4]^-}$

(B) $\mathrm{[Cr(CN)_6]^{3-}}$

(C) $\mathrm{[NiF_6]^{2-}}$

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(D) CrO_2Cl_2

Q.17

H_2S acts only as a reducing agent while SO_2 can act both as a reducing and oxidizing agent because

(A) oxygen is more $-$ ve in SO_2

(B) hydrogen in H_2S is more $+$ ve than oxygen

(C) S in SO_2 has one oxidation state

(D) S in H_2S has -2 oxidation state

Q.18

Point out the correct statement of the following about $Na_2S_4O_6$

(A) Average oxidation number of S atom is $+2$

(B) Oxidation number of two S atoms is zero each and that of other two is $+5$ each

(C) Oxidation number of two S atoms is $+1$ each and that of other two is $+4$ each

(D) None of these

Q.19

Standard reduction potentials of the half reactions are given below:

$F_2(g) + 2e^- \rightarrow 2F^-(aq); E^\circ = +2.85\ V$

$Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq); E^\circ = +1.36\ V$

$Br_2(l) + 2e^- \rightarrow 2Br^-(aq); E^\circ = +1.06\ V$

$I_2(s) + 2e^- \rightarrow 2I^-(aq); E^\circ = +0.53\ V$

The strongest oxidising and reducing agents respectively are:

(A) F_2 and I^-

(B) Br_2 and Cl^-

(C) Cl_2 and Br^-

(D) Cl_2 and I_2

Q.20

A mixture of potassium chlorate, oxalic acid and sulphuric acid is heated. During the reaction which element undergoes maximum change in the oxidation number?

(A) S

(B) H

(C) Cl

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(D) C

Answers & Solutions

Q.1 Answer:

$Zn + 2HCl \longrightarrow ZnCl_2 + H_2$

Solution:

(a) and (b) are neutralisation reactions. The oxidation state of Cu **is** $+2$ **in both reactant and product, and** SO_4^{2-} **ion does not change. In option (d) is a redox reaction.**

$Zn \rightarrow Zn^{2+} + 2e^-$ **(Oxidation)**

$2H^+ + 2e^- \rightarrow H_2$ **(Reduction)**

Q.2 Answer:

NO_3^- is reduced in preference to hydronium ion

Solution:

Zinc gives H_2 **gas with dil** H_2SO_4/HCl **but not with** HNO_3 **because in** HNO_3, NO_3^- **ion is reduced and give** $NH_4NO_3,\ N_2O, NO$ **and** NO_2 **(based upon the concentration of** HNO_3**)**

$[Zn + 2HNO_3 \longrightarrow Zn(NO_3)_2 + 2H] \times 4$

(nearly 6\%)

$$HNO_3 + 8H \longrightarrow NH_3 + 3H_2O$$

$NH_3 + HNO_3 \longrightarrow NH_4NO_3$

$4Zn + 10HNO_3 \longrightarrow$

$$4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$$

Zn **is on the top position of hydrogen in electrochemical series. So** Zn **displaces** H_2 **from dilute** H_2SO_4 **and** HCl **with liberation of** H_2**.**

$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

Q.3 Answer:

$NaH + H_2O \longrightarrow NaOH + H_2$

Solution:

(a) and (b) are disproportionation reactions.

(c) is an oxidation reaction.

$2H^- \longrightarrow H_2 + 2e^-$

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Q.4 Answer:

II, III,IV

Solution:

II, III and IV are redox reactions.

Q.5 Answer:

16.89

Solution:

$ClO_3^- \longrightarrow Cl_2^0$

$x - 6 = -1 \quad x = 0$

$x = +5 \quad x = 0 (x = \textbf{oxidation number})$

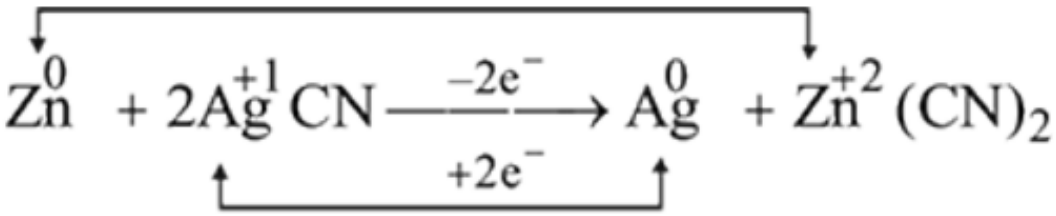
Equivalent mass $= \frac{\text{Molecular mass}}{\text{Oxidation number}}$

$= \frac{84.45}{5} = 16.89$

Q.6 Answer:

$Zn + 2AgCN \rightarrow 2Ag + Zn(CN)_2$

Solution:



The oxidation state shows a change only in (d)

Q.7 Answer:

F_2 and ClO_4^-

Solution:

F_2 being most electronegative element cannot exhibit any positive oxidation state. In ClO_4^- chlorine is present in its highest oxidation state i.e $+7$. Therefore it does not show disproportionation reaction.

Q.8 Answer:

$MgCO_3 \longrightarrow MgO + CO_2$

Solution:

Intermolecular redox reaction and redox reactions are same

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Q.9 Answer:

2.5

Solution:

$\mathrm{Na_2\ S_4O_6 : 2 + 4x - 12 = 0}$

$\therefore \mathrm{x} = \frac{10}{4} = 2.5$

Oxidation number of $\mathrm{S} = 2.5$

Q.10 Answer:

$+2$

Solution:

Let X **be the oxidation number of** Fe **in** $\mathrm{Na_2\ [Fe(CN)_5NO]}$.

The Na atoms have oxidation number of $+1$.

The $\mathrm{NO+}$ **group has oxidation number of 1.**

The CN **group has oxidation number of** -1.

$2(+1) + X + 5(-1) + 1 = 0$

$2 + X - 5 + 1 = 0$

$X - 2 = 0$

$X = +2$

The oxidation number of Fe **in** $\mathrm{Na_2\ [Fe(CN)_5NO]}$ **is** $+2$.

Q.11 Answer:

$16/3$

Solution:

$\mathrm{Fe_3O_8}$

$3x - 16 = 0 \Rightarrow x = \frac{16}{3}$

Q.12 Answer:

$+2$

Solution:

$$\begin{bmatrix} +2 & 0 & 3- \\ \mathrm{Pt} & (\mathrm{C_2H_4}) & \mathrm{Cl_3} \end{bmatrix}$$

Oxidation number of $\mathrm{Pt} = +2$

Q.13 Answer:

(i), (ii) and (iii)

Solution:

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(i) $Mn^{n+} + ne^- \rightleftharpoons M$, **for this reaction, high negative value of** E° **indicates lower reduction potential, that means** M **will be a good reducing agent.**

$$\left[\begin{array}{c} \text{Stronger reducing agent} \Rightarrow \text{Easy to oxidise} \\ \Downarrow \\ \text{Lower reduction potential} \Leftarrow \text{higher oxidation potential} \end{array}\right]$$

(ii)

Element	F	Cl	Br	I
Reduction potential (E° volt)	+2.87	+1.36	+1.06	+0.54

As reduction potential decreases from fluorine to iodine, oxidising nature also decreases from fluorine to iodine.

(iii) The size of halide ions increases from F^- **to** I^-**. The bigger ion can loose electron easily. Hence the reducing nature increases from** HF **to HI.**

Q.14 Answer:

$3, 1, 6$

Solution:

On balancing the given reaction, we find $3Na_2HAsO_3 + NaBrO_3 + 6HCl$

$$\longrightarrow 6NaCl + 3H_3AsO_4 + NaBr$$

Q.15 Answer:

$S_2O_4^{2-} < SO_3^{2-} < S_2O_6^{2-}$

Solution:

In SO_3

$x + 3(-2) = -2; x = +4$

In $S_2O_4^{--}$

$2x + 4(-2) = -2$

$2x - 8 = -2$

$2x = 6; \quad x = +3$

In $S_2O_6^{2-}$

$2x + 6(-2) = -2$

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$2x = 10; \quad x = +5$

hence the correct order is $S_2O_4^{--} < SO_3^{--} < S_2O_6^{-}$

Q.16 Answer:

CrO_2Cl_2

Solution:

CrO_2Cl_2

Let O. No. of $Cr = x$

$\therefore \quad x + 2(-2) + 2(-1) = 0$

$x - 4 - 2 = 0$

$\therefore \quad x = +6$

Q.17 Answer:

S in H_2S has -2 oxidation state

Solution:

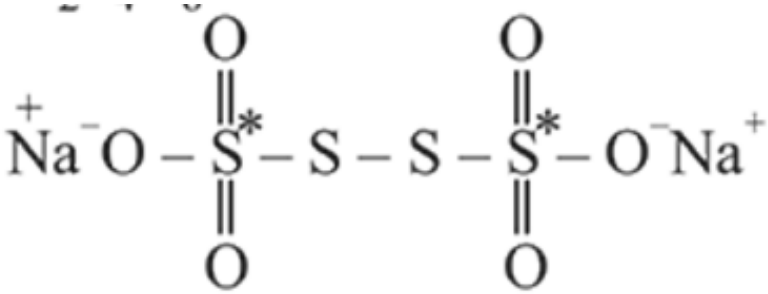
H_2S, the oxidation state of S is -2. So it cannot accept more electrons because on accepting 2 electrons S accquires a noble gas configuration. So, it can acts only as a reducing agent by loosing electron. On the other hand, the oxidation state of S in SO_2 is $+4$ which is an intermediate oxidation state of sulphur so it can reduce as well oxidise.

Q.18 Answer:

Oxidation number of two S atoms is zero each and that of other two is $+5$ each

Solution:

$Na_2S_4O_6$ has the structure :



O.N. of two S^* atoms are $+5$ each and that of other two S atoms is zero each.

Q.19 Answer:

F_2 and I^-

Solution:

Higher the value of reduction potential higher will be the oxidising power whereas the lower the

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value of reduction potential higher will be the reducing power.

Q.20 Answer:

Cl

Solution:

$$\mathrm{KClO_3 + H_2C_2O_4 + H_2^{+6}SO_4 \rightarrow K_2^{+6}SO_4 + KCl + CO_2 + H_2O}$$

i.e. maximum change in oxidation number is observed in $\mathrm{Cl}(+5 \text{ to } -1)$

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